Alex Anthony

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Data Scientist | Python | Machine Learning | A/B Testing | Forecasting | SQL | MSc Data Science

PROFILE

Data analyst with 1.5 years of commercial experience across asset management, marketing analytics and food delivery, building Python ML models, data pipelines and reporting used by senior leadership. Independent projects cover Bayesian marketing mix modelling, A/B testing and experimentation, quantitative trading research and ML forecasting, each with a live dashboard and public code here.

TECHNICAL SKILLS

ML / Data Science: Bayesian modelling (PyMC), marketing mix modelling (adstock, saturation, ROI), budget optimisation, A/B testing and experiment design (power analysis, SRM checks, delta method, CUPED/CUPAC, sequential testing, multiple-testing correction), causal inference and uplift modelling, time-series forecasting and backtesting, XGBoost, SHAP, Optuna, K-means segmentation, cohort analysis, cointegration, Kalman filter, NLP (VADER), hypothesis testing

Development: Python (pandas, NumPy, scikit-learn, SciPy, PyMC, FastAPI, SQLAlchemy, Streamlit, Plotly, pytest), SQL (CTEs, window functions, query optimisation), PostgreSQL/TimescaleDB, Docker, Git, cron automation, R, HTML/CSS, Claude Code

BI & Data: Power BI (DAX, Power Query, star-schema modelling), Tableau, advanced Excel (XLOOKUP, Power Pivot), GA4, Monday.com API, Power Automate, ETL pipelines, data validation and reconciliation, MI and financial reporting

Domain: FCA-regulated asset management and fund data, marketing measurement and campaign analytics, stakeholder communication, translating business questions into analysis, data quality and governance, documentation

WORK EXPERIENCE

Marketing Data Analyst - Fixed-Term Contract *Apr 2026 - Present*

Marlborough Group Holdings - FCA-regulated asset management (Remote)

- Gave senior marketing, sales and investment leadership their first quantitative view of client acquisition across individual, intermediary and institutional channels by building Python ML segmentation models and ETL pipelines end to end
- Made fund and client reporting audit-ready and reproducible with Python and SQL pipelines that extract, validate and reconcile data across multiple financial sources
- Eliminated a recurring multi-hour manual process by rebuilding the UK sales sector pipeline in Python, expanding coverage from 4 to 56 sectors with ISIN-matched validation
- Gave leadership cross-channel visibility with GA4 and Monday.com API-connected dashboards, and let the team query procedure documents in plain language through a SharePoint Copilot agent

Data Analyst Intern *Nov 2025 - Jan 2026*

Uptrail Ltd, London

- Maintained and enhanced Power BI dashboards for marketing and business stakeholders, ensuring consistent, up-to-date visibility into campaign and performance metrics that supported weekly decision-making
- Extracted and validated data using SQL, producing clean, reliable datasets that underpinned stakeholder reporting and removed the need for manual cross-checking
- Built Power Automate workflows to eliminate manual handoffs across reporting processes, freeing team capacity for higher-value analytical work
- Produced management information and financial performance reports for senior stakeholders, with data quality controls embedded at every stage

Data Scientist *Apr 2024 - Sept 2024*

Zomato, Gurugram, India

- Shaped re-engagement and campaign strategy by identifying high-value and high-churn customers through Python segmentation and cohort analysis of 50,000+ orders
- Improved operational planning across 12 delivery zones with regression and time-series demand forecasts, and replaced a manual weekly KPI collation with daily SQL pipelines

KEY PROJECTS

Adstock - Bayesian Marketing Mix Model & Budget Optimiser (Python, PyMC, SciPy, Streamlit)

- Found that the largest channel (62% of budget) returned under £1 per £1, and that reallocating the same budget could add about £587k revenue per quarter (97% probability of beating the current mix), using adstock and saturation curves fitted in PyMC
- Validated the attribution, not just the fit: in a parameter-recovery test the true ROI fell inside the 90% interval in 93% of cases, and a 26-week holdout forecast scored 4.4% MAPE against 22% for a naive baseline | **Live Dashboard** | **GitHub**

Holdout - A/B Testing & Experimentation Analysis (Python, SciPy, scikit-learn, Streamlit)

- Recommended the winning email from a 64,000-customer randomised test (+118% spend per customer) after SRM and balance checks, delta-method intervals and Holm correction, and advised against deploying an uplift model that did not beat the simple policy on held-out customers
- Showed by simulation that checking results daily raises false positives from 4% to 24%, and built an mSPRT sequential test that keeps them under 5%, plus CUPED variance reduction and a power and sample-size planner | **Live Dashboard** | **GitHub**

Cointegrated - UK Equity Pairs-Trading Research Platform (Python, TimescaleDB, Docker, Streamlit)

- Showed that a UK pairs-trading strategy's edge is not distinguishable from luck (deflated Sharpe 41%) and disappears when trading costs double, using out-of-sample testing that never touched the pair-selection data
- Traced one pair's entire loss to a single corporate event (May 2024) and documented the missing stop-loss as a limitation
- Replaced manual data refreshes with an automated daily pipeline (Python, TimescaleDB, Docker, cron) that feeds an append-only paper-trading record and a live Streamlit dashboard | **Live Dashboard** | **GitHub**

Automated Factsheet Checking Platform - Marlborough Group Holdings (Python, FastAPI)

- Replaced a compliance-critical manual check across 15 partner fund ranges with a documented, repeatable platform that validates data against acceptance criteria, flags discrepancies and produces Excel audit output

Crypto Volatility Forecasting - MSc Dissertation, 78% Distinction (XGBoost, SHAP, Optuna, VADER)

- Built an end-to-end forecasting system on 132,109 financial records combining XGBoost with sentiment signals, and found and fixed a critical data integrity issue before deployment | **Live Dashboard** | **GitHub**

INDEPENDENT POWER BI DASHBOARDS

Bank Loan Analysis Dashboard | Retail Sales & Commercial Performance Dashboard - star-schema modelling, DAX and commercial analytics

EDUCATION

MSc Data Science - University of Greenwich, London (2:1, dissertation 78% Distinction) *2024 - 2025*

BSc Data Science & Analytics - Sharda University, India (8.59 GPA) *2021 - 2024*

- Co-authored a research paper comparing K-means, DBSCAN and hierarchical clustering on 18,185 records

ADDITIONAL INFORMATION

Languages: English (fluent) | Interests: swimming, fitness | Open to relocation